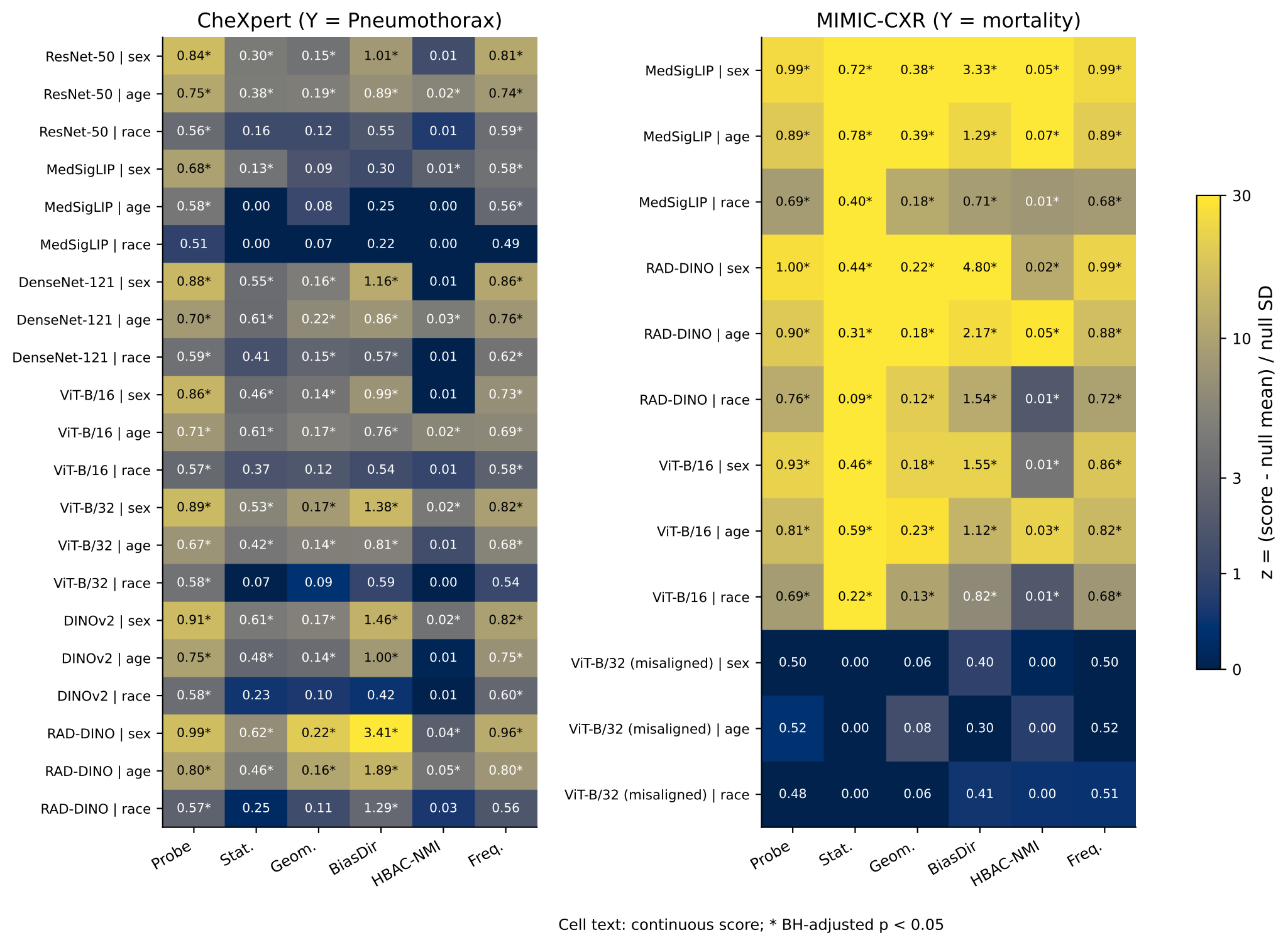
CheXpert (Y = Pneumothorax)
ResNet-50 | sex 0.84* 0.30* 0.15* 1.01* 0.01 0.81*
ResNet-50 | age 0.75* 0.38* 0.19* 0.89* 0.02* 0.74*
ResNet-50 | race 0.56* 0.16 0.12 0.55 0.01 0.59*
MedSigLIP | sex 0.68* 0.13* 0.09 0.30 0.01* 0.58*
MedSigLIP | age 0.58* 0.00 0.08 0.25 0.00 0.56*
MedSigLIP | race 0.51 0.00 0.07 0.22 0.00 0.49
DenseNet-121 | sex 0.88* 0.55* 0.16* 1.16* 0.01 0.86*
DenseNet-121 | age 0.70* 0.61* 0.22* 0.86* 0.03* 0.76*
DenseNet-121 | race 0.59* 0.41 0.15* 0.57* 0.01 0.62*
ViT-B/16 | sex 0.86* 0.46* 0.14* 0.99* 0.01 0.73*
ViT-B/16 | age 0.71* 0.61* 0.17* 0.76* 0.02* 0.69*
ViT-B/16 | race 0.57* 0.37 0.12 0.54 0.01 0.58*
ViT-B/32 | sex 0.89* 0.53* 0.17* 1.38* 0.02* 0.82*
ViT-B/32 | age 0.67* 0.42* 0.14* 0.81* 0.01 0.68*
ViT-B/32 | race 0.58* 0.07 0.09 0.59 0.00 0.54
DINOv2 | sex 0.91* 0.61* 0.17* 1.46* 0.02* 0.82*
DINOv2 | age 0.75* 0.48* 0.14* 1.00* 0.01 0.75*
DINOv2 | race 0.58* 0.23 0.10 0.42 0.01 0.60*
RAD-DINO | sex 0.99* 0.62* 0.22* 3.41* 0.04* 0.96*
RAD-DINO | age 0.80* 0.46* 0.16* 1.89* 0.05* 0.80*
RAD-DINO | race 0.57* 0.25 0.11 1.29* 0.03 0.56
Probe
Stat.
Geom.
BiasDir
HBAC-NMI
Freq.
MIMIC-CXR (Y = mortality)
MedSigLIP | sex 0.99* 0.72* 0.38* 3.33* 0.05* 0.99*
MedSigLIP | age 0.89* 0.78* 0.39* 1.29* 0.07* 0.89*
MedSigLIP | race 0.69* 0.40* 0.18* 0.71* 0.01* 0.68*
RAD-DINO | sex 1.00* 0.44* 0.22* 4.80* 0.02* 0.99*
RAD-DINO | age 0.90* 0.31* 0.18* 2.17* 0.05* 0.88*
RAD-DINO | race 0.76* 0.09* 0.12* 1.54* 0.01* 0.72*
ViT-B/16 | sex 0.93* 0.46* 0.18* 1.55* 0.01* 0.86*
ViT-B/16 | age 0.81* 0.59* 0.23* 1.12* 0.03* 0.82*
ViT-B/16 | race 0.69* 0.22* 0.13* 0.82* 0.01* 0.68*
ViT-B/32 (misaligned) | sex 0.50 0.00 0.06 0.40 0.00 0.50
ViT-B/32 (misaligned) | age 0.52 0.00 0.08 0.30 0.00 0.52
ViT-B/32 (misaligned) | race 0.48 0.00 0.06 0.41 0.00 0.51
Probe
Stat.
Geom.
BiasDir
HBAC-NMI
Freq.
z = (score - null mean) / null SD
30
10
3
1
0
Cell text: continuous score; * BH-adjusted p < 0.05